SPECIFICATION

EDGE-COMPUTING SYSTEM AND METHOD FOR MONITORING THE ACCUMULATION OF FLOATING SOLID WASTE IN A WATERWAY

TECHNICAL FIELD OF THE INVENTION

The present invention relates to environmental monitoring, and more particularly to a system and a method in which a camera fixed above a creek, river or drainage channel supplies images to an edge computing device that detects, tracks and counts floating solid waste inside an operator-defined region of the water surface and raises an alert when the waste has accumulated to a set level.

BACKGROUND OF THE INVENTION

Creeks and other urban waterways act as drainage channels for densely populated areas. Bottles, bags and food containers discarded into drains collect on the water surface where the current slows, narrow the channel and contribute to flooding when heavy rain arrives. Local government units commonly learn of such a build-up only when personnel walk along the bank, which is slow, labor-intensive and periodic, and which produces no dated record of how the build-up developed.

Several approaches to the problem are known. Image-recognition methods for garbage, such as that of Chinese Patent No. CN111767822B, identify garbage in pictures of monitored rivers and similar areas. Water-surface cleaning units such as that of Korean Patent No. KR101985436B1 locate floating trash with a deep-learning model in order to collect it mechanically. United States Patent No. US11308603B2 discloses detection and monitoring of waste disposal using artificial intelligence on images from sensors or cameras. United States Patent No. US11788877B2 monitors waste placed in bins and containers with load cells and cameras. Flood warning systems such as that of United States Patent No. US12094320B2 raise alerts from water-level and rainfall sensors.

These approaches leave several needs unmet. Detectors that classify individual images do not determine how much waste has built up over time at a fixed place, and a raw per-frame count fluctuates with every missed or false detection. A count taken over the whole image also includes banks, walls, bridges and floating barriers that lie within the camera's view. When a high-resolution

camera picture is reduced to the input size of a detector running on a low-power computer, small floating items become too small to detect. Sensor-based alerts respond to water level rather than to the waste that causes the blockage, and container-based systems are not suited to moving water. Finally, a system that stops when mains power fails cannot be relied upon at a creek during the storms when monitoring matters most.

SUMMARY OF THE INVENTION

It is an object of the present invention to provide a self-contained system that is installed beside a waterway and that reports when floating solid waste in a defined stretch of the water surface has accumulated to a level that requires clearing.

It is another object of the invention to count only the waste lying on the water within a region drawn by the operator, so that fixed objects in the camera's view are not counted.

It is a further object of the invention to keep small floating items detectable on a low-power edge computing device by passing only the part of each high-resolution frame that contains the monitored region to the detector.

It is still a further object of the invention to raise one alert for each build-up, without repeated alerts while the amount of waste hovers near the set level, and to keep a dated record of each alert with photographic evidence.

It is yet another object of the invention to keep the camera and the edge computing device operating from a common battery during a failure of the mains supply.

To achieve these objects, the invention provides a system comprising a network camera mounted above the water surface and an edge computing device that receives the camera's video stream. The edge computing device stores a monitored region in the form of a polygon drawn by the operator over the image of the water; crops each frame to a padded bounding box of the polygon before passing it to a convolutional neural network object detector trained to detect bottles, bags and food containers; assigns each detected item a track identity that persists from frame to frame; counts only items whose bounding-box center lies inside the polygon; takes the median of the counts of the most recent frames as a smoothed count; raises a single accumulation alert when the smoothed count

reaches a threshold; and re-arms the alert only after the smoothed count has fallen to a re-arm level below the threshold and a minimum interval has elapsed. On each alert the system sounds an alarm, shows the alert on a display, sends a notification to a mobile device and stores a record with the class counts and an annotated snapshot. The camera and the edge computing device are powered from a common uninterruptible power supply module.

The invention further provides a corresponding method of monitoring the accumulation of floating solid waste in a waterway.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an isometric view of the exterior of the system according to an embodiment of the invention, showing the monitoring unit with its door closed, the network camera on its base and the cables that leave the monitoring unit.

FIG. 2 is an isometric view of the interior of the monitoring unit of FIG. 1, with the door removed and the right side wall and the top wall shown in phantom lines.

FIG. 3 is a flowchart of the method according to an embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

Referring to FIG. 1, the system comprises a network camera (36) and a monitoring unit housed in an enclosure (10), and may further comprise a cloud relay and a mobile device, which are not shown in the drawings. The network camera (36) is a fixed outdoor camera with a weatherproof rating, carried on a camera base (38) and mounted above the water surface of a waterway so that its field of view covers the stretch of water to be monitored, in which floating solid waste items collect. In the embodiment, the camera is a bullet network camera of 4-megapixel resolution rated IP67, configured to deliver a main video stream of 1280 x 720 pixels at 15 frames per second over the Real-Time Streaming Protocol, and a lower-resolution sub-stream of 640 x 360 pixels that serves as a fallback.

The enclosure (10) is a metal junction box with outside dimensions, in the embodiment, of about 250 mm in width, 300 mm in height and 150 mm in depth. A touchscreen display (16), in the embodiment a 7-inch panel of 1024 x 600 pixels,

is set in a cut-out in the door (12) of the enclosure, so that it can be read and touched without opening the enclosure. The door (12) is hinged along its left edge by hinges (14) and carries a handle with a lock (18) below the display. A case fan (20) is fitted in each side wall behind a wire guard. Three cable entries (24) are cut low in the right side wall and sealed with rubber sealant. From front to back they carry a USB extension cable (26), a mains cable (28) and a composite cable (30) that runs to the camera (36). A wireless network adapter (32) with two antennas (34) sits on top of the enclosure, outside the metal walls, so that its signal is not shielded, and is connected to the inside of the enclosure by the USB extension cable (26).

Referring to FIG. 2, the enclosure (10) contains an edge computing device (50). The edge computing device (50) is a single-board computer with a quad-core ARM processor and 8 GB of memory, fitted with an active cooler (52). It is connected to the camera (36) through its wired network port by the composite cable (30), which carries both the network signal and direct-current power to the camera; in the embodiment a cable of 40 m was used. A solid-state drive (56), carried on a PCI Express adapter board (54) mounted above the edge computing device, holds the detection model, the event database and the snapshots, while a high-endurance memory card in the edge computing device holds the operating system, so that the continuous writing of records does not wear the operating system's storage.

The edge computing device (50) is powered by an uninterruptible power supply module (47) mounted beneath it, which carries rechargeable lithium-ion cells (48), in the embodiment four cells of the 21700 size. The adapter board (54), the edge computing device (50) and the power supply module (47) form a board stack that stands on a mounting plate (46) raised on standoffs above the floor of the enclosure. An alternating-current power strip (62) lies on the floor of the enclosure beneath the mounting plate (46) and receives the mains cable (28). A power adapter (40) plugged into the power strip (62) charges the power supply module (47). The power supply module (47) also has a USB output, to which a USB hub (42) is connected; a DC-DC step-up converter (44) connected to the hub raises the 5 V USB supply to the 12 V required by the camera and feeds the power conductors of the composite cable (30). The camera (36) and the edge computing device (50) therefore draw from the same battery-backed supply, so

that detection continues during a failure of the mains supply. The edge computing device reads the battery state from the power supply module and, when the remaining charge falls below a set level, begins a timed countdown and shuts down cleanly, so that no record is cut off mid-write.

The touchscreen display (16) is connected to the edge computing device (50) by display cables (60) for video and touch input and shows a dashboard with the live annotated video, the current count against the threshold, the count of each class and the alert state. An audible alarm (58), in the embodiment a passive buzzer mounted on the back wall of the enclosure and driven through buzzer wires (64) by pulse-width modulation from a general-purpose input/output pin, sounds when an alert is raised. The two case fans (20) draw air through the enclosure and are powered through fan cables (66) from USB ports of the power strip (62), so that they do not draw on the battery. The wireless network adapter (32) connects the edge computing device to a local wireless network, which leaves the wired port of the edge computing device dedicated to the camera link.

In use, the camera (36) is fixed to a mount, such as a fence or railing beside the waterway, at a height above the water surface, and is aimed along the surface so that its field of view covers the stretch to be monitored. The enclosure (10) is placed under a shelter, such as a barangay building, and is joined to the camera only by the composite cable (30). Only the camera is exposed to the weather. The height of the camera affects the size at which floating items appear in the image. In field tests of the embodiment, the proportion of items detected and the accuracy of the class labels both increased as the camera was lowered from about 4.0 m (13 ft) to about 2.7 m (9 ft) and to about 2.1 m (7 ft) above the water.

The method carried out by the edge computing device (50) is described with reference to FIG. 3. On start-up (100) the edge computing device loads the detection model, the stored monitored region and the threshold, and connects to the camera's main stream. In the capturing step (102), a separate reading thread keeps only the newest decoded frame, so that the detector never works on a backlog of old frames; if no frame arrives for a set period the camera is reconnected, and if the main stream repeatedly gives no picture the system switches to the sub-stream.

The monitored region is a polygon of at least three points that the operator draws by tapping points along the waterline on the live image shown on the

touchscreen display (16) or on the mobile device. The points are stored as fractions of the frame width and height, so that the region remains valid if the stream resolution changes, and the region is reloaded after every restart. The region is drawn after the camera has been fixed in its final position, since it is traced over the camera's actual view.

In the cropping step (104), the edge computing device computes the bounding box of the monitored region in the frame, enlarges it by a margin of 4 percent of the frame dimensions, enlarges it further if necessary so that each side is at least 320 pixels, and keeps it inside the frame, giving a crop box. Only the part of the frame inside the crop box is passed to the detector. When the crop box would cover 95 percent or more of the frame, cropping is skipped because it would save no processing time. Because the detector resizes its input to a fixed size, cropping before resizing leaves floating items larger in the detector's input than resizing the whole frame would.

In the detecting and tracking step (106), a convolutional neural network object detector, in the embodiment a single-stage detector of the nano size exported to an inference format optimized for ARM processors and run at an input size of 960 pixels, detects items of the three classes of bottle, bag and food container in the cropped image and returns for each a bounding box, a class and a confidence score. The boxes are mapped back to the coordinates of the full frame. The detector is trained on images of floating waste that include images taken at the deployment site, and is evaluated on a held-out test set that is fixed once and not used for training. A multi-object tracker then assigns each detection a track identity that persists while the item remains in view, so that the same item keeps the same box and label from frame to frame. For display, the tracker's identities are replaced by small numbers starting at 1; a number is released once its item has been missing for about thirty frames.

In the counting step (108), the center of each bounding box is tested against the polygon of the monitored region. Detections whose center lies inside the polygon are counted by class; detections whose center lies outside the polygon, such as items on the bank, on a bridge or on a floating barrier, are drawn in a muted style and are not counted. The count of the frame is the number of counted detections.

In the smoothing step (110), the edge computing device keeps the per-frame counts of the most recent N frames, in the embodiment fifteen frames or about three seconds of video, and takes their median as the smoothed count. A single missed or false detection therefore does not change the smoothed count. No alert is raised until N frames have been collected after start-up.

In the decision step (112), when no alert is active and the smoothed count reaches the threshold, and at least a minimum interval, in the embodiment 60 seconds, has passed since the previous alert, the accumulation alert is raised (114). The alert remains active, without being raised again, until the smoothed count falls to or below the re-arm level (120), at which point the alert is cleared (122) and the system is ready to raise the next alert. The re-arm level is the threshold minus one quarter of the threshold rounded down, and at least one item below the threshold; for a threshold of 20 items it is 15, and for a threshold of 15 it is 12. The difference between the two levels prevents a pile of waste that hovers near the threshold from raising a new alert every few seconds. The threshold has a default of 20 items and can be changed from the touchscreen display or the mobile device, so that it can be set to suit each site.

When the per-frame count reaches the threshold for at least three consecutive frames but the smoothed count does not reach it, the event is recorded as a suppressed spike, together with its number of frames and its peak count, so that brief surges can be reviewed without raising an alert.

When the alert is raised (114), the touchscreen display (16) shows an alert banner over the live image and the counter is shown in an alert color; the audible alarm (58) repeats until the alert is muted, except during quiet hours set by the operator; and the alert is sent to the mobile device. When the mobile device is on the same local network, it reads the state directly from a web server on the edge computing device. When it is elsewhere, the edge computing device uploads its state, the event records and compressed snapshots to the cloud relay, a real-time database, and reads from it the commands left by the mobile device, such as muting the alert or changing the threshold. Uploads are made less frequent when no mobile device is watching. The mobile device shows a notification with the item count, the threshold and the snapshot.

In the storing step (116), an event record is written to a local relational database on the solid-state drive (56). The record contains the date and time, the

count of each class, the peak confidence, the threshold in use and the path of a snapshot. The snapshot is the frame with the boxes, the monitored region and an information card drawn on it; an unannotated copy of the same frame is saved beside it, so that the evidence for each alert can later be labelled and used to retrain the detector. Manual captures, test alerts, mutes and power events are recorded in the same way, while frames that raise no alert are displayed but not stored, which keeps the database small during a long deployment. A summary of the processing latency is also written every 100 frames. The results are then shown on the dashboard (118), and the method returns to the capturing step (102) for the next frame.

The threshold is preferably calibrated at the site. In the embodiment, the count shown on the dashboard was compared at intervals with a count of the items made by eye within the monitored region at the same moment, and the threshold was then set so that the alert is raised at the amount of waste at which the operator wants to act.

In a test of the embodiment, the detector reached a precision of 84.6 percent, a recall of 67.2 percent and a mean average precision at an overlap threshold of 0.5 of 0.744 on a fixed test set of 807 images. At a creek in Barangay 870, Pandacan, Manila, the system operated for 24 hours over six days without a crash or overheating, processed each frame in an average of 275 to 350 milliseconds, raised the alert in each of ten trials, and delivered the notification to a mobile device on the local network within about 0.1 to 2 seconds. Over the three deployment days, 295 of 395 class labels were correct (74.7 percent), rising from 62.5 percent with the camera at about 4.0 m to 73.5 percent at about 2.7 m and 85.0 percent at about 2.1 m.

The invention is not limited to the embodiment described. Another camera, single-board computer, detector architecture, set of waste classes, threshold, number of frames for smoothing or minimum interval may be used without departing from the scope of the claims, and the cloud relay and mobile device may be omitted where only local alerting is needed.

INDUSTRIAL APPLICABILITY

The system and method can be manufactured from commercially available components and used by local government units, barangays, drainage authorities

and waste collection services to know when a stretch of a creek, river or drainage channel needs clearing, to schedule cleanup from a dated record of build-ups, and to keep photographic evidence of each event.

CLAIMS

What is claimed is:

1. A system for monitoring the accumulation of floating solid waste in a waterway, comprising a network camera (36) mounted above a water surface of the waterway; an edge computing device (50) connected to the camera (36) to receive its video stream and configured to run a convolutional neural network object detector trained to detect floating solid waste; a display (16); an audible alarm (58); and a non-volatile storage (56), characterized in that the edge computing device (50) is configured to:
   a. store a monitored region in the form of a polygon drawn by an operator over an image of the water surface;
   b. crop each frame of the video stream to a crop box that encloses the monitored region with a margin, pass the cropped image to the object detector, and map the resulting detections back to the coordinates of the frame;
   c. assign each detection a track identity that persists from frame to frame;
   d. count, in each frame, only the detections whose bounding-box center lies inside the polygon of the monitored region;
   e. compute a smoothed count as the median of the counts of a set number of most recent frames;
   f. raise a single accumulation alert when the smoothed count reaches a threshold, and re-arm the alert only after the smoothed count has fallen to a re-arm level below the threshold and a minimum interval has elapsed since the previous alert; and
   g. upon raising the accumulation alert, sound the audible alarm (58), show the alert on the display (16), and store in the non-volatile storage (56) an event record comprising the date and time, the count of each class, the threshold in use and an annotated snapshot of the frame.
2. The system according to claim 1, wherein the set number of frames is fifteen, the minimum interval is sixty seconds, and the re-arm level is the threshold minus one quarter of the threshold rounded down.
3. The system according to claim 1 or claim 2, wherein the margin of the crop box is a fraction of the frame dimensions, each side of the crop box is

enlarged to at least a minimum number of pixels, and cropping is skipped when the crop box covers at least 95 percent of the frame.

4. The system according to any one of claims 1 to 3, wherein the edge computing device (50) records a suppressed spike when the count of a frame reaches the threshold for at least three consecutive frames without the smoothed count reaching the threshold.

5. The system according to any one of claims 1 to 4, wherein the edge computing device (50) stores, beside each annotated snapshot, an unannotated copy of the same frame for use in retraining the object detector.

6. The system according to any one of claims 1 to 5, further comprising an uninterruptible power supply module (47) with rechargeable cells (48) that powers the edge computing device (50) and, through a USB hub (42) and a DC-DC step-up converter (44), the network camera (36), such that the camera and the edge computing device continue to operate from a common battery during a failure of the mains supply, the edge computing device being configured to shut down cleanly when the remaining charge falls below a set level.

7. The system according to any one of claims 1 to 6, wherein the edge computing device (50), the uninterruptible power supply module (47) and the non-volatile storage (56) are contained in an enclosure (10) placed under a shelter, the display (16) is a touchscreen set in a cut-out in a door (12) of the enclosure, the enclosure has case fans (20) and sealed cable entries (24), and the network camera (36) is connected to the edge computing device by a composite cable (30) that carries both the network signal and power.

8. The system according to any one of claims 1 to 7, further comprising a mobile device and a cloud relay, wherein the edge computing device (50) uploads its state, the event records and compressed snapshots to the cloud relay and reads from it commands left by the mobile device, including a command to mute the alert and a command to change the threshold, and wherein the mobile device shows a notification with the item count, the threshold and the snapshot.

9. The system according to any one of claims 1 to 8, wherein the object detector detects the classes of bottle, bag and food container, the threshold has a

default of twenty items, and the network camera (36) is mounted at a height of not more than about 3 m above the water surface.

10. A method of monitoring the accumulation of floating solid waste in a waterway using the system according to claim 1, comprising the steps of:

    a. fixing the network camera (36) above the water surface and, with the camera in its final position, drawing the monitored region as a polygon over the image of the water surface;

    b. reading frames from the video stream of the camera (102);

    c. cropping each frame to the crop box enclosing the monitored region (104) and detecting floating solid waste in the cropped image with the convolutional neural network object detector (106);

    d. assigning each detection a persistent track identity (106);

    e. counting only the detections whose bounding-box center lies inside the monitored region (108);

    f. computing the smoothed count as the median of the counts of the most recent frames (110);

    g. raising a single accumulation alert when the smoothed count reaches the threshold (112, 114) and re-arming it only after the smoothed count has fallen to the re-arm level and the minimum interval has elapsed (120, 122); and

    h. storing an event record with an annotated snapshot (116) and notifying a mobile device of the alert.

11. The method according to claim 10, further comprising calibrating the threshold at the site by comparing the count shown by the system with a count made by eye of the items within the monitored region at the same moment.

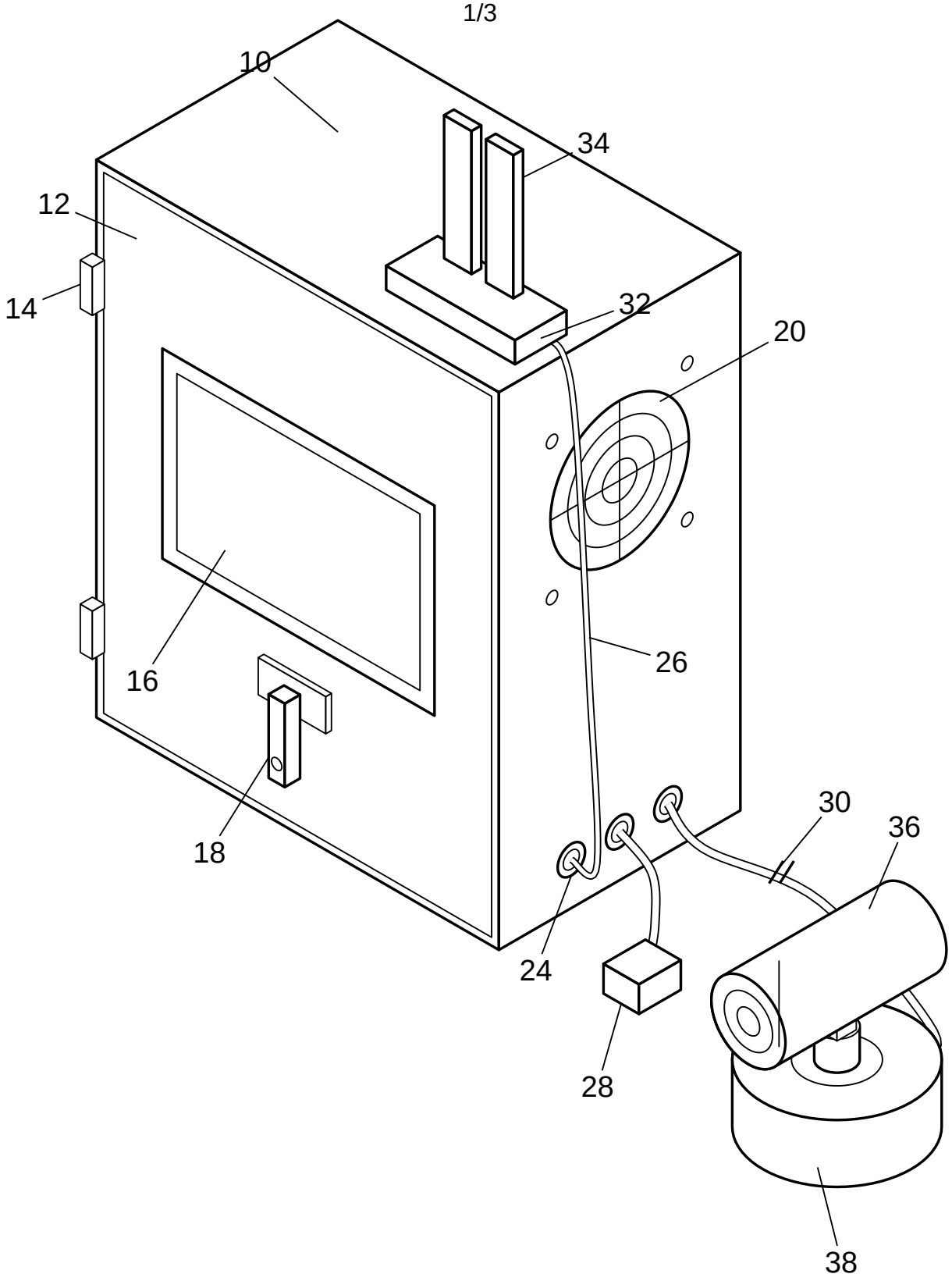


FIG. 1

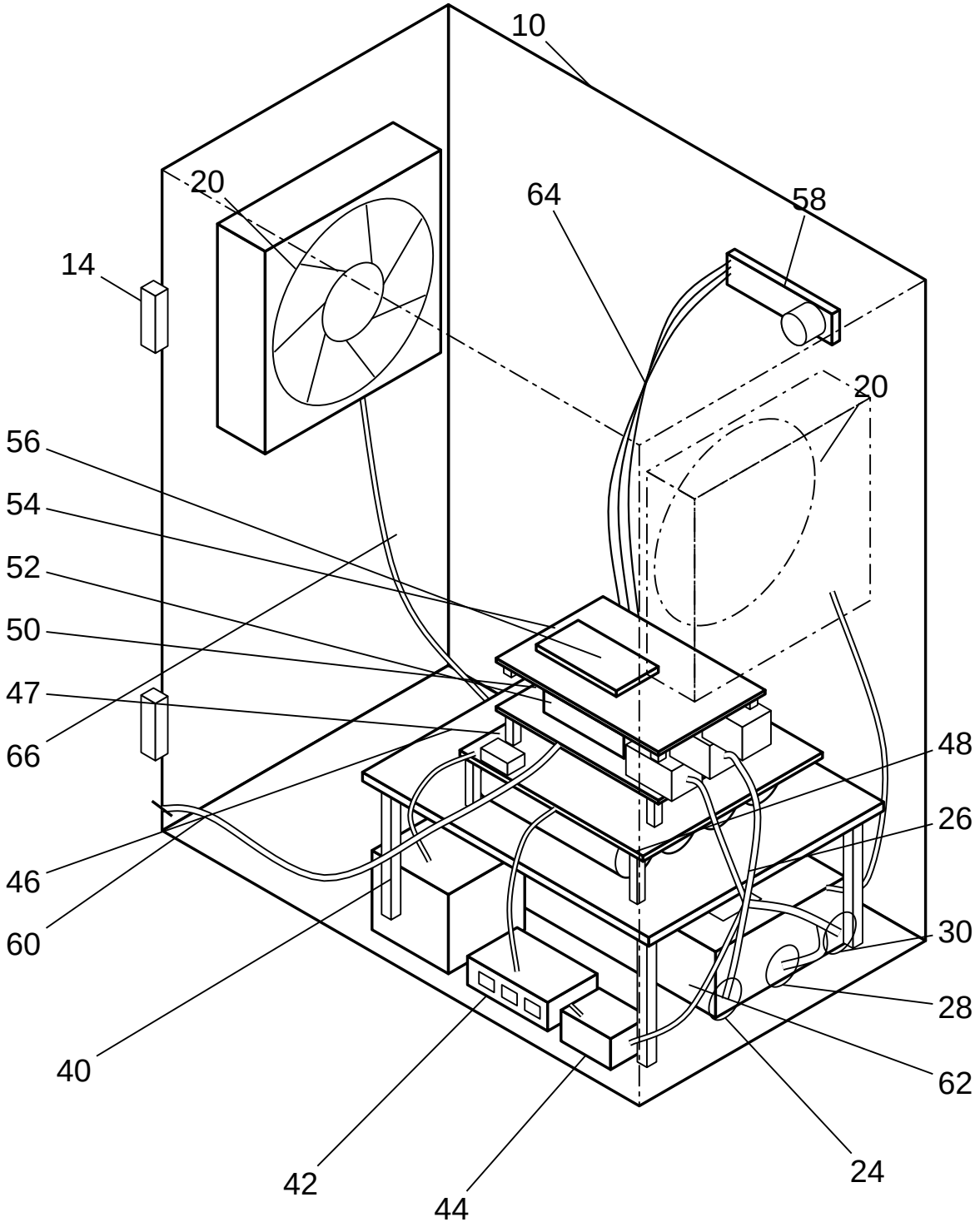


FIG. 2

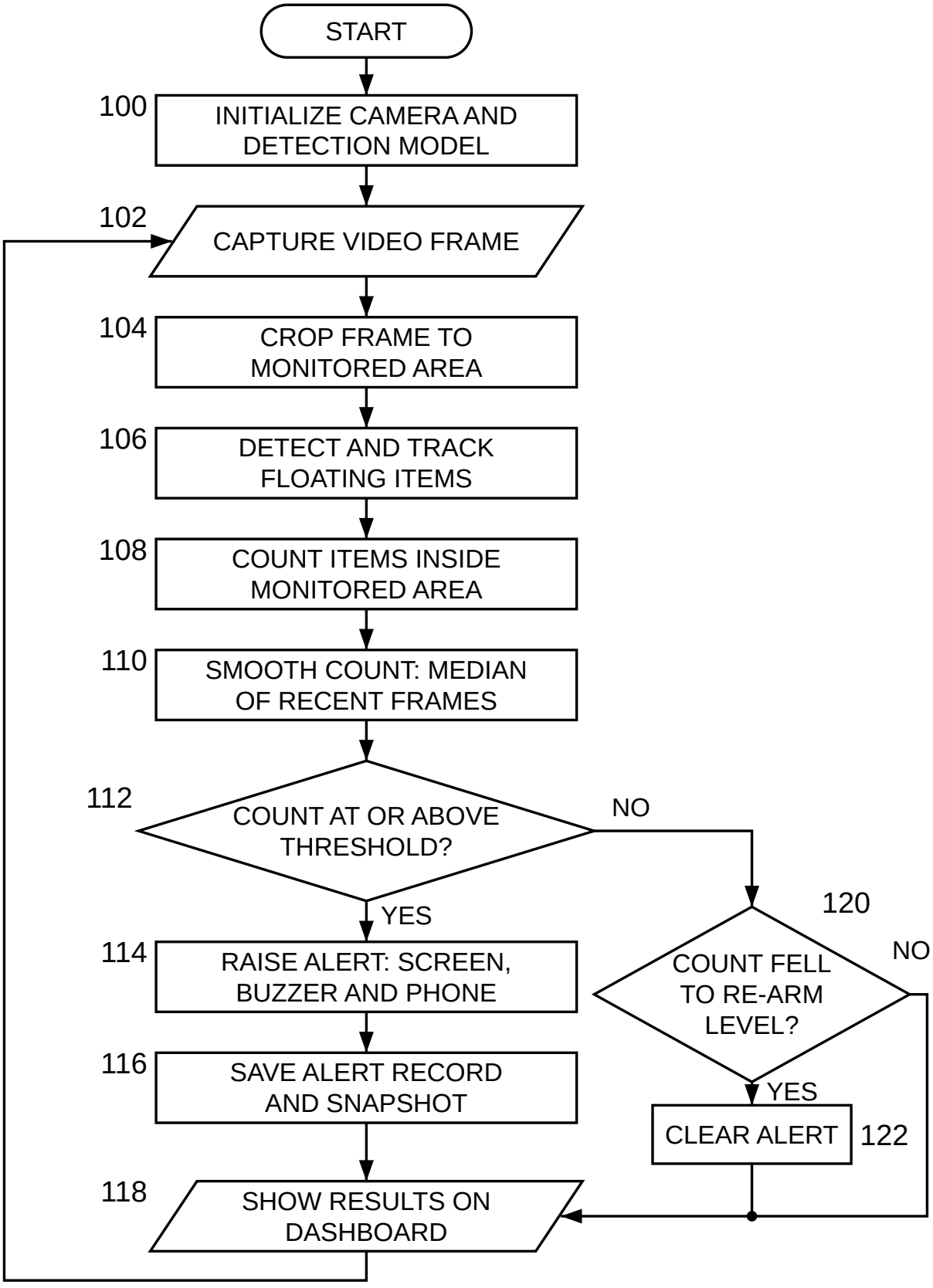

START
100
INITIALIZE CAMERA AND DETECTION MODEL
102
CAPTURE VIDEO FRAME
104
CROP FRAME TO MONITORED AREA
106
DETECT AND TRACK FLOATING ITEMS
108
COUNT ITEMS INSIDE MONITORED AREA
110
SMOOTH COUNT: MEDIAN OF RECENT FRAMES
112
COUNT AT OR ABOVE THRESHOLD?
NO
YES
114
RAISE ALERT: SCREEN, BUZZER AND PHONE
116
SAVE ALERT RECORD AND SNAPSHOT
118
SHOW RESULTS ON DASHBOARD
120
COUNT FELL TO RE-ARM LEVEL?
NO
YES
CLEAR ALERT
122

FIG. 3

ABSTRACT

EDGE-COMPUTING SYSTEM AND METHOD FOR MONITORING THE ACCUMULATION OF FLOATING SOLID WASTE IN A WATERWAY

A system and method for monitoring the accumulation of floating solid waste in a waterway. A network camera (36) above the water surface supplies video to an edge computing device (50). The device crops each frame to a padded box around an operator-drawn polygon, detects bottles, bags and food containers with a convolutional neural network, tracks each item across frames, and counts only items whose box center lies inside the polygon. The median count of recent frames is compared with a threshold: one alert is raised when the threshold is reached, and the alert re-arms only below a lower level after a minimum interval. Each alert sounds an alarm (58), appears on a touchscreen (16), is stored with an annotated snapshot and is sent to a mobile device. An uninterruptible power supply (47) powers both camera and computer.